

Controls HW 3

Sina Hazeghi

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1 Unicycle Differential Flatness

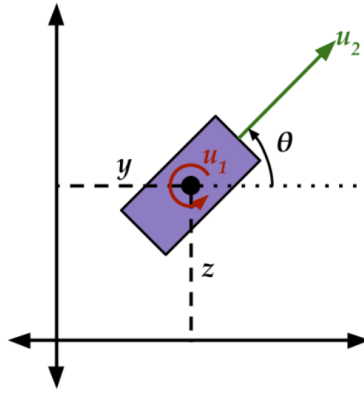


Figure 1: Unicycle system

Have unicycle system with dynamics:

$$q = x = \begin{bmatrix} y \\ z \\ \theta \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \dot{x} = f(x, u) = \begin{bmatrix} \cos(\theta)u_2 \\ \sin(\theta)u_2 \\ u_1 \end{bmatrix}$$

Note that dynamics are first order since controls are directly first order

a) Compute Differential Flatness

We should find "flat outputs" of unicycle system s.t given outputs(t) we can compute state and controls at time t as a function of these outputs and their time derivatives.

Consider y,z flat outputs, we can use them to compute theta:

$$\frac{\dot{z}}{\dot{y}} = \frac{\cos \theta u_2}{\sin \theta u_2} \implies \theta = \arctan\left(\frac{\dot{z}}{\dot{y}}\right) \forall u_2 \neq 0$$

Since $u_1 = \dot{\theta}$:

$$u_1 = \frac{d}{dt}\theta = \frac{d}{dt} \arctan\left(\frac{\dot{z}}{\dot{y}}\right) = \frac{\ddot{z}\dot{y} - \dot{z}\ddot{y}}{\dot{y}^2 + \dot{z}^2}$$

again defined when $u_2 \neq 0$

Finally we can compute u_2 as a function of y or z (use z here):

$$u_2 = \frac{\dot{z}}{\sin \theta} = \sqrt{\dot{y}^2 + \dot{z}^2}$$

Some conditions to use these equations:

$$\dot{y} > 0, u_2 > 0$$

b) Creating a non-colliding Trajectory

We are tasked with creating a trajectory that takes our unicycle system from $x_0 \rightarrow x_f$, while avoiding randomly placed circular obstacles.

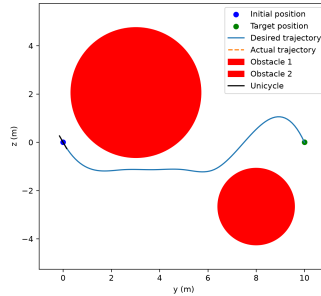
Quick Sketch of methodology:

- Define N way points linearly spaced between start and goal

$$\{(t_0, y_0, z_0), (t_1, y_1, z_1), \dots, (t_N = t_f, y_N = y_f, z_N = z_f)\}$$

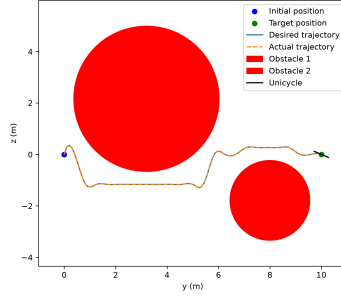
- Iterate over way points, if any of them within radius r of obstacle, move them above or below that obstacle
 - As long as we N is large enough, interpolation between points also won't hit the obstacle
- Use cubic spline interpolation among the waypoints

Using this algorithm we ensure that we won't hit any obstacles, y is always increasing so $\dot{y} > 0$ and the trajectory is twice differentiable (need 2nd derivatives to compute u_1). Below is an example trajectory we have created:



Once we have created the desired trajectory in the flat outputs (task space), we can easily use our equations from part a) to compute the necessary controls to follow this trajectory. Since we have controls + initial state we just integrate forward the dynamics in time to follow our trajectory.

Below you can see that the desired trajectory can be followed exactly, despite our underactuated system:



Discrete Time LQR and Stability

a) Deriving Algebraic Ricatti

Have HJB in the form:

$$V(x_k) = \min_{u_k} (x_k^\top Q x_k + u_k^\top R u_k + V(x_{k+1}))$$

$$x_{k+1} = A x_k + B u_k, \quad V(x) = x^\top S x$$

$$\text{RHS} = \min_{u_k} (x_k^\top Q x_k + u_k^\top R u_k + (A x_k + B u_k)^\top S (A x_k + B u_k))$$

Expanding the last term:

$$\begin{aligned} (A x_k + B u_k)^\top S (A x_k + B u_k) &= x_k^\top A^\top S A x_k + x_k^\top A^\top S B u_k \\ &\quad + u_k^\top B^\top S A x_k + u_k^\top B^\top S B u_k \end{aligned}$$

Now take the derivative and set to zero:

$$\begin{aligned} \frac{\partial}{\partial u_k} (\dots) &= 2 R u_k + B^\top S A x_k + B^\top S A x_k + 2 B^\top S B u_k \\ &= 2 ((R + B^\top S B) u_k + B^\top S A x_k) = 0 \end{aligned}$$

$$(R + B^\top S B) u_k = -B^\top S A x_k \implies u_k^* = -(R + B^\top S B)^{-1} B^\top S A x_k$$

Substituting into the dynamics:

$$x_{k+1} = A x_k + B u_k^* = (A - B(R + B^\top S B)^{-1} B^\top S A) x_k$$

And back into the Bellman equation:

$$\begin{aligned} x_k^\top S x_k &= x_k^\top Q x_k + x_k^\top A^\top S B (R + B^\top S B)^{-1} R (R + B^\top S B)^{-1} B^\top S A x_k \\ &\quad + x_k^\top (-A^\top S B (R + B^\top S B)^{-1} B^\top + A^\top) S (A - B (R + B^\top S B)^{-1} B^\top S A) x_k \end{aligned}$$

Since this holds for all x_k and both sides are symmetric:

$$\begin{aligned} S &= Q + A^\top S B (R + B^\top S B)^{-1} R (R + B^\top S B)^{-1} B^\top S A \\ &\quad + (-A^\top S B (R + B^\top S B)^{-1} B^\top + A^\top) S (A - B (R + B^\top S B)^{-1} B^\top S A) \end{aligned}$$

Letting $K = (R + B^\top S B)^{-1} B^\top S A$:

$$S = Q + K^\top R K + (-K^\top B^\top + A^\top) S (A - B K) = Q + K^\top R K + (A - B K)^\top S (A - B K)$$

Letting $K = (R + B^\top S B)^{-1} B^\top S A$:

$$S = Q + K^\top R K + (-K^\top B^\top + A^\top) S (A - B K)$$

Expanding the last term:

$$(A - B K)^\top S (A - B K) = A^\top S A - A^\top S B K - K^\top B^\top S A + K^\top B^\top S B K$$

Note $K^\top R K$ and $K^\top B^\top S B K$ share the factors $K^\top$ and K , so they group as $K^\top (R + B^\top S B) K$. From the stationarity condition, $(R + B^\top S B) K = B^\top S A$, hence

$$K^\top (R + B^\top S B) K = K^\top B^\top S A,$$

which cancels exactly against the cross term $-K^\top B^\top S A$:

$$\begin{aligned} S &= Q + K^\top \underbrace{(R + B^\top S B) K}_{= B^\top S A} - A^\top S B K - K^\top B^\top S A + A^\top S A \\ &= Q + A^\top S A - A^\top S B K \end{aligned}$$

Substituting K back in gives the discrete-time algebraic Riccati equation:

$$\boxed{S = Q + A^\top S A - A^\top S B (R + B^\top S B)^{-1} B^\top S A}$$

b) Lyapunov Stability for Discrete-Time LQR

Theorem. If $\exists V$ such that

$$V(0) = 0, \quad V(x) > 0 \quad \forall x \neq 0,$$

$$V(x_{k+1}) - V(x_k) < 0 \quad \forall x_k \neq 0,$$

then $x = 0$ is asymptotically stable.

Take $V^*(x_k) = x_k^\top S x_k$ from LQR, with S the stabilizing solution of the DARE. Then

$$V^*(0) = 0^\top S 0 = 0, \quad V^*(x) = x^\top S x > 0 \quad \forall x \neq 0 \quad (\text{since } S \succ 0).$$

And via the HJB equation,

$$V^*(x_k) = g(x_k, u^*) + V^*(x_{k+1}) \implies V^*(x_{k+1}) - V^*(x_k) = -g(x_k, u^*),$$

where $g(x_k, u_k) = x_k^\top Q x_k + u_k^\top R u_k$ is the running cost. Since $Q \succ 0$ and $R \succ 0$,

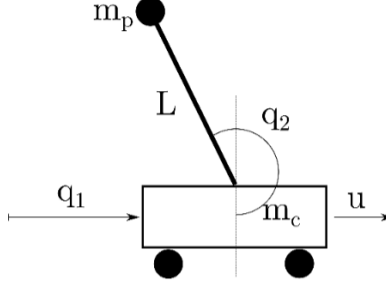
$$g(x_k, u^*) = x_k^\top Q x_k + u^{*\top} R u^* > 0 \quad \forall x_k \neq 0,$$

and therefore

$$V^*(x_{k+1}) - V^*(x_k) < 0 \quad \forall x_k \neq 0.$$

Thus V^* satisfies both conditions of the Lyapunov stability theorem, so $x = 0$ is asymptotically stable under $u_k = -Kx_k$.

2 Energy Shaping the Cartpole System



We have cartpole system with dynamics:

$$M(q)\ddot{q} + C(q, \dot{q}) = Bu$$

Where for constants $m_c = m_p = L = 1$ and $g = 9.81$ We have:

$$\begin{pmatrix} 2 & \cos q_2 \\ \cos q_2 & 1 \end{pmatrix} \ddot{q} + \begin{pmatrix} -\dot{q}_2^2 \sin q_2 \\ g \sin q_2 \end{pmatrix} = \begin{pmatrix} u \\ 0 \end{pmatrix}$$

This gives us the equations:

$$2\ddot{q}_1 + \ddot{q}_2 \cos q_2 - \dot{q}_2 \sin q_2 = u \tag{1}$$

$$\ddot{q}_1 \cos q_2 + \ddot{q}_2 + g \sin q_2 \tag{2}$$

a) Reparametrizing Underactuated Controls

Since our system is underactuated and we really only have control over $\ddot{q}_1$ directly via our control u , we'd like to reparametrize our equations s.t. we can get desired u in terms of only desired $\ddot{q}_{1,d}$ and current state $\mathbf{x} = (q, \dot{q})^T$. From 2nd equation of motion rewrite $\ddot{q}_2$ in terms of $\ddot{q}_{1,d}$ and state:

$$\ddot{q}_2 = -\ddot{q}_{1,d} \cos q_2 - g \sin q_2$$

So after substituting this back into eqn (1) and applying trig identity:

$$u(\ddot{q}_{1,d}, \mathbf{x}) = \ddot{q}_{1,d}(1 + \sin^2 q_2) - g \sin q_2 \cos q_2 - \dot{q}_2^2 \sin q_2$$

b) Proving our input is Energy-Shaping

For this section for simplicity we will refer to (q_1, q_2) as (y, θ) where y is horizontal position and θ is angle of pendulum. We define the energy of the pendulum portion of the system:

$$E(\mathbf{x}) = \frac{1}{2}\dot{\theta}^2 - g \cos \theta$$

This is just the sum of KE + PE at any state $\mathbf{x}$. To keep our pendulum upright, we would like our energy to exactly be the PE of the pendulum with $\cos \theta = 1$ so we have $E^* = g$ (0 velocity). We consider now our "distance" from desired energy term:

$$\hat{E} = E - E^*$$

We would like this term to go to 0, from either direction so we want $|\hat{E}| \rightarrow 0$. Consider now the lyapunov-like function, the mean squared error of our energy

$$V(E) = \frac{1}{2}(E - E^*)^2 = \frac{1}{2}\hat{E}^2$$

Since $V(E) \geq 0 \forall E$ if we can show that $\exists u$ s.t. $\dot{V} < 0$ until we get to $V(0) = 0$ then we know that the L1 norm $|\hat{E}| \rightarrow 0$ as well and that we can stabilize the cartpole s.t. the pendulum portion is completely upright.

$$\begin{aligned} \frac{dV}{dt} &= \frac{dV}{d\hat{E}} \frac{d\hat{E}}{dt} = \hat{E} \cdot \dot{\hat{E}} \\ \dot{\hat{E}} &= \dot{E} - \dot{E}^* = \dot{E} \end{aligned}$$

Recall that for any given $\ddot{y}_d$ and state $\mathbf{x}$ we can compute the necessary control u to achieve this and in part a) we reparametrized also $\ddot{\theta}$ in terms of $\ddot{y}_d$. Consider the energy shaping input:

$$\ddot{y}_d = \dot{\theta} \cos \theta \hat{E}$$

Then we have by a):

$$\ddot{\theta} = -\ddot{y}_d \cos \theta - g \sin \theta = -\dot{\theta} \hat{E} \cos^2 \theta - g \sin \theta$$

And plugging this into $\dot{\hat{E}}$, the $g \sin \theta$ term cancels:

$$\dot{\hat{E}} = -\dot{\theta}^2 \cos^2 \theta \hat{E}$$

And therefore we have:

$$\dot{V} = -(\dot{\theta} \cos \theta \hat{E})^2 \leq 0$$

and this is < 0 whenever $\dot{\theta} \neq 0$ and our angle is not upright and our goal $\hat{E}$ hasn't yet reached 0 so V is non-increasing $\implies |\hat{E}|$ is also non-increasing

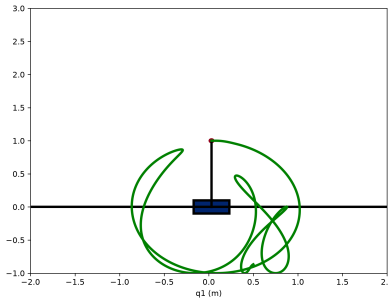
c) Implementation

We implement our energy shaping controller, adding additional PD terms to $\ddot{y}_d$ s.t. it forces the cartpole to not only balance the pendulum portion but also move and stabilize towards the origin:

$$\ddot{y}_d = \dot{\theta} \cos \theta \hat{E} - k_p y - k_d \dot{y}$$

Once we we get "close enough", defined by the magnitude of $|\hat{E}|$ as well as the angular distance (i.e. how far is $|\theta, \dot{\theta}|$ from $(\pi, 0)$) then we switch from our energy shaping controller to LQR to stabilize the pendulum.

I found that the gains $k_p = 5, k_d = 5$ worked and let me achieve the following result, which as you can see both stabilizes the pendulum at $\theta = \pi$ as well as keeps $y = 0$



Lyapunov Stability

Consider $\dot{x} = f(x)$.

WTS: If $\exists$ a Lyapunov function V , then $x = 0$ is lyapunov stable.

Assume $\exists V$, continuously differentiable, such that

$$V(0) = 0, \quad V(x) > 0 \quad \forall x \in \mathbb{R}^n \setminus \{0\}, \quad \dot{V}(x) \leq 0.$$

Given $\varepsilon > 0$, consider $B_\varepsilon = \{x \in \mathbb{R}^n : \|x\| < \varepsilon\}$.

(a)

Let $\alpha = \min_{\|x\|=\varepsilon} V(x)$. Show $\alpha > 0$.

Consider the set $S = \{x \in \mathbb{R}^n : \|x\| = \varepsilon\}$. For all $x \in S$,

$$\|x\| = \varepsilon > 0 \implies x \neq 0 \implies V(x) > 0.$$

Therefore $\alpha = \min_{x \in S} V(x) > 0$.

(b)

Consider the sublevel set of the Lyapunov function intersected with B_ε :

$$\Omega_\beta = \{x : V(x) \leq \beta\} \cap B_\varepsilon, \quad \beta \in (0, \alpha).$$

WTS $\Omega_\beta \subseteq \text{int}(B_\varepsilon)$.

By construction $\Omega_\beta \subseteq B_\varepsilon$. Moreover, by definition of $\beta \in (0, \alpha)$, there is no $x \in \Omega_\beta$ such that $\|x\| = \varepsilon$: indeed

$$x \in \Omega_\beta \implies V(x) \leq \beta < \alpha,$$

whereas $\alpha = \min_{x \in S} V(x)$ with $S = \{x : \|x\| = \varepsilon\}$. So if x was on the boundary of B_ε , i.e. $x \in S$ then it wouldn't be possible for it also to be true that $V(x) < \alpha$ therefore we know that $\Omega_\beta \subseteq \text{Int}(B_\varepsilon)$

(c)

Now consider any trajectory $x(t)$ s.t. $x(0) \in \Omega_\beta$, then since we have that $\dot{V}(x) < 0 \forall x \neq 0$ then for any $t > 0$ any time during the trajectory

$$V(x(t)) < V(x(0)) \leq \beta \implies x(t) \in \Omega_\beta \forall t > 0$$

Therefore Ω_β is positively invariant

(d)

By V continuous and $V(0) = 0$ we know $\exists \delta > 0$ s.t:

$$B_\delta = \{x : \|x\| < \delta\} \subset \Omega_\beta$$

i.e. there's some ball of inputs x around 0 s.t. $V(x) \leq \beta$ Take some trajectory s.t. $x(0) \in B_\delta$ then since $B_\delta \subset \Omega_\beta$ and Ω_β positive invariant then we know that:

$$x(0) \in B_\delta \implies x(t) \in \Omega_\beta \implies x(t) \in \text{Int}(B_\varepsilon) \forall t > 0$$

This is literally saying that $x(t)$ is lyapunov stable, recall defn of lyapunov stability: For any $\epsilon > 0 \exists \delta > 0$ s.t.

$$\|x(0)\| < \delta \implies \|x(t)\| < \epsilon \forall t > 0$$

I.e. if for any trajectory, given the lyapunov function exists, if we start in the smaller B_δ we will never leave the larger ball B_ϵ (ϵ is given and we pick δ s.t. this is true)